ANDROID DEVELOPMENT SERIES

Multiple Android Activities

Engineering Intent-Based Navigation and Lifecycle Management

STUDENT	**Student Name**
ASSIGNMENT	**03: Multi-Activity Architecture**
REPOSITORY	**github.com/fizzflip/rivi**

3.1 Application Architecture Context

- **Contextual Application:** Rivi (Minimalist Academic Microlearning)
- **Architectural Driver:** Model-View-ViewModel (MVVM) Design Pattern
- **Core Features Elaborated:** Real-world educational applications heavily rely on segregating user workflows into distinct spatial sandboxes. By distributing the user-flow across multiple `Activities`, the application naturally inherits built-in back-stack management handled by the Android OS.

In Rivi, we focus on engineering the explicit mapping between a global navigation dashboard and hyper-focused functional interfaces, bridging them through the strategic dispatch of `Intents` bearing bundled payload data (such as primitive subject identification keys).

3.2 Core Blueprint: AndroidManifest.xml

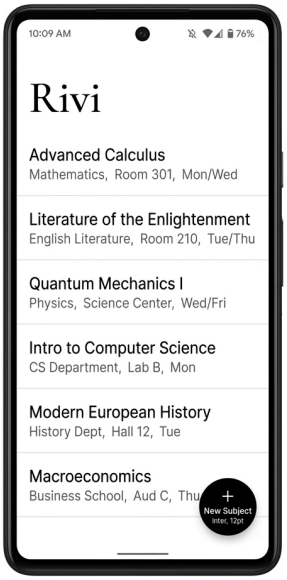
The manifest serves as the directory for all entry points. Each activity must be registered here to be accessible via Intent dispatching.

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Rivi">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".ui.StudySessionActivity"
            android:exported="false"
            android:theme="@style/Theme.Rivi" />
        <activity
            android:name=".ui.CardManagementActivity"
            android:exported="false"
            android:theme="@style/Theme.Rivi" />
        <activity
            android:name=".ui.SessionRecapActivity"
            android:exported="false"
            android:theme="@style/Theme.Rivi" />
    </application>

</manifest>
```

3.3 Activity 1: Dashboard (MainActivity)

The `MainActivity` operates as the primary entry gateway. It is responsible for orchestrating global user progression by fetching subjects and allowing navigation to study sessions or management views.

- **Functional Logic:** It observes a Room database via a ViewModel. When a category is clicked, it synthesizes an `Explicit Intent`, attaches a `SubjectID`, and invokes `startActivity()`.



3.3.1 Implementation Code

```
package me.prasad.study_app;

import android.app.DatePickerDialog;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.animation.AnimationUtils;
import android.view.animation.LayoutAnimationController;
import android.widget.ImageButton;
import android.widget.Toast;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.lifecycle.ViewModelProvider;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.floatingactionbutton.ExtendedFloatingActionButton;
import com.google.android.material.textfield.TextInputEditText;

import java.text.SimpleDateFormat;
import java.util.Calendar;
```

```java
import java.util.Locale;

import me.prasad.study_app.data.entity.Subject;
import me.prasad.study_app.ui.CardManagementActivity;
import me.prasad.study_app.ui.StudySessionActivity;
import me.prasad.study_app.ui.adapter.SubjectAdapter;
import me.prasad.study_app.viewmodel.SubjectViewModel;

public class MainActivity extends AppCompatActivity {

    private final Calendar calendar = Calendar.getInstance();
    private SubjectViewModel viewModel;
    private final ActivityResultLauncher<String> importLauncher = registerForActivityResult(
            new ActivityResultContracts.GetContent(),
            uri -> {
                if (uri != null) {
                    importSubject(uri);
                }
            }
    );
    private SubjectAdapter adapter;
    private int pendingExportSubjectId = -1;
    private final ActivityResultLauncher<String> exportLauncher = registerForActivityResult(
            new ActivityResultContracts.CreateDocument("application/json"),
            uri -> {
                if (uri != null) {
                    exportSubject(uri);
                }
            }
    );

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        setupRecyclerView();

        viewModel = new ViewModelProvider(this).get(SubjectViewModel.class);
        viewModel.getAllSubjects().observe(this, subjects -> adapter.submitList(subjects));

        ExtendedFloatingActionButton fab = findViewById(R.id.fab_add_subject);
        fab.setOnClickListener(v -> showAddSubjectDialog());

        ImageButton btnImport = findViewById(R.id.btn_import);
        btnImport.setOnClickListener(v -> importLauncher.launch("application/json"));
    }

    private void importSubject(Uri uri) {
        try {
           viewModel.importSubject(getContentResolver().openInputStream(uri), () -> runOnUiThread(()
-> Toast.makeText(this, "Subject imported", Toast.LENGTH_SHORT).show()));
        } catch (Exception e) {
            Toast.makeText(this, "Import failed", Toast.LENGTH_SHORT).show();
        }
    }

    private void exportSubject(Uri uri) {
        if (pendingExportSubjectId == -1) return;
        try {
         viewModel.exportSubject(pendingExportSubjectId, getContentResolver().openOutputStream(uri));
            Toast.makeText(this, "Exporting...", Toast.LENGTH_SHORT).show();
```

```
            pendingExportSubjectId = -1;
        } catch (Exception e) {
            Toast.makeText(this, "Export failed", Toast.LENGTH_SHORT).show();
        }
    }

    private void showAddSubjectDialog() {
        View view = LayoutInflater.from(this).inflate(R.layout.dialog_add_subject, null);
        TextInputEditText nameInput = view.findViewById(R.id.edit_subject_name);
        TextInputEditText dateInput = view.findViewById(R.id.edit_exam_date);
        MaterialButton btnSave = view.findViewById(R.id.btn_save);
        MaterialButton btnCancel = view.findViewById(R.id.btn_cancel);

        AlertDialog dialog = new AlertDialog.Builder(this)
                .setView(view)
                .create();

        dateInput.setOnClickListener(v -> new DatePickerDialog(this, (view1, year, month, dayOfMonth)
-> {
            calendar.set(Calendar.YEAR, year);
            calendar.set(Calendar.MONTH, month);
            calendar.set(Calendar.DAY_OF_MONTH, dayOfMonth);
            SimpleDateFormat sdf = new SimpleDateFormat("MMM dd, yyyy", Locale.getDefault());
            dateInput.setText(sdf.format(calendar.getTime()));
                                    },    calendar.get(Calendar.YEAR),    calendar.get(Calendar.MONTH),
calendar.get(Calendar.DAY_OF_MONTH)).show());

        btnSave.setOnClickListener(v -> {
            if (nameInput.getText() == null || dateInput.getText() == null) return;

            String name = nameInput.getText().toString().trim();
            if (name.isEmpty() || dateInput.getText().toString().isEmpty()) {
                Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show();
                return;
            }

            Subject subject = new Subject(name, calendar.getTimeInMillis(), 0);
            viewModel.insert(subject);
            dialog.dismiss();
        });

        btnCancel.setOnClickListener(v -> dialog.dismiss());

        dialog.show();
    }

    private void setupRecyclerView() {
        RecyclerView recyclerView = findViewById(R.id.recycler_subjects);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));

                  LayoutAnimationController  animation  =  AnimationUtils.loadLayoutAnimation(this,
R.anim.layout_slide_up);
        recyclerView.setLayoutAnimation(animation);
        adapter = new SubjectAdapter(new ViewModelProvider(this).get(SubjectViewModel.class), this);
        recyclerView.setAdapter(adapter);

        adapter.setOnSubjectClickListener(new SubjectAdapter.OnSubjectClickListener() {
            @Override
            public void onSubjectClick(Subject subject) {
                Intent intent = new Intent(MainActivity.this, StudySessionActivity.class);
                intent.putExtra(StudySessionActivity.EXTRA_SUBJECT_ID, subject.getSubjectId());
                startActivity(intent);
```

```
            }

            @Override
            public void onManageCardsClick(Subject subject) {
                Intent intent = new Intent(MainActivity.this, CardManagementActivity.class);
                intent.putExtra(CardManagementActivity.EXTRA_SUBJECT_ID, subject.getSubjectId());
                startActivity(intent);
            }

            @Override
            public void onExportClick(Subject subject) {
                pendingExportSubjectId = subject.getSubjectId();
                String name = subject.getName();
               String fileName = (name != null ? name.replaceAll("\\s+", "_") : "subject") + ".json";
                exportLauncher.launch(fileName);
            }
        });
    }
}
```

3.3.2 Layout Declaration

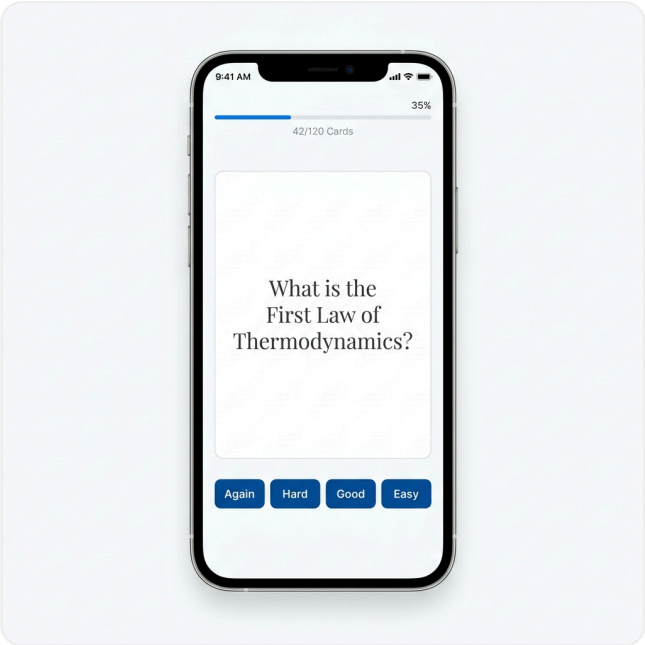
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout    xmlns:android="http://schemas.android.com/apk/
res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/text_app_title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/app_name"
        android:textAppearance="@style/TextAppearance.Rivi.Academic"
        android:textSize="32sp"
        android:textStyle="bold"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <ImageButton
        android:id="@+id/btn_import"
        style="@style/Widget.Rivi.IconButton"
        android:layout_width="48dp"
        android:layout_height="48dp"
        android:contentDescription="@string/import_subject"
        android:src="@drawable/ic_file_upload"
        app:layout_constraintBottom_toBottomOf="@id/text_app_title"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toTopOf="@id/text_app_title"
        app:tint="@color/rivi_primary" />

    <TextView
        android:id="@+id/text_app_subtitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/app_subtitle"
        android:textAppearance="@style/TextAppearance.Rivi.UI"
        android:textColor="@color/rivi_text_secondary"
```

```
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/text_app_title" />

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recycler_subjects"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_marginTop="32dp"
        android:clipToPadding="false"
        android:paddingBottom="88dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/text_app_subtitle"
        tools:listitem="@layout/item_subject" />

    <com.google.android.material.floatingactionbutton.ExtendedFloatingActionButton
        android:id="@+id/fab_add_subject"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        android:text="@string/new_subject"
        android:textColor="@color/white"
        app:backgroundTint="@color/black"
        app:icon="@drawable/ic_add"
        app:iconTint="@color/white"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

3.4 Activity 2: Flashcard Study (StudySessionActivity)

This activity facilitates the core “Active Recall” protocol. It abandons lists for a high-intensity, distraction-free card interface.

- **Functional Logic:** It extracts the `SubjectID` from the incoming intent, queries the specific flashcard set, and manages user feedback (Again, Hard, Good, Easy) back to the SR algorithm.



3.4.1 Implementation Code

```
package me.prasad.study_app.ui;

import android.animation.AnimatorInflater;
import android.animation.AnimatorSet;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.LinearLayout;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
import androidx.lifecycle.ViewModelProvider;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.card.MaterialCardView;

import java.util.List;

import me.prasad.study_app.R;
import me.prasad.study_app.data.entity.Flashcard;
import me.prasad.study_app.data.entity.Subject;
import me.prasad.study_app.logic.SRSLogic;
import me.prasad.study_app.viewmodel.FlashcardViewModel;
import me.prasad.study_app.viewmodel.SubjectViewModel;

public class StudySessionActivity extends AppCompatActivity {
```

```java
    public static final String EXTRA_SUBJECT_ID = "extra_subject_id";

    private FlashcardViewModel flashcardViewModel;

    private TextView textSubjectTitle, textSessionProgress, textCardContent;
    private MaterialCardView cardFlashcard;
    private LinearLayout layoutGrading;

    private List<Flashcard> sessionCards;
    private Subject currentSubject;
    private boolean isShowingAnswer = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_study_session);

        int subjectId = getIntent().getIntExtra(EXTRA_SUBJECT_ID, -1);
        if (subjectId == -1) {
            finish();
            return;
        }

        initViews();
        setupViewModels(subjectId);
    }

    private void initViews() {
        textSubjectTitle = findViewById(R.id.text_subject_title);
        textSessionProgress = findViewById(R.id.text_session_progress);
        textCardContent = findViewById(R.id.text_card_content);
        cardFlashcard = findViewById(R.id.card_flashcard);
        layoutGrading = findViewById(R.id.layout_grading);

        MaterialButton btnAgain = findViewById(R.id.btn_again);
        MaterialButton btnHard = findViewById(R.id.btn_hard);
        MaterialButton btnGood = findViewById(R.id.btn_good);
        MaterialButton btnEasy = findViewById(R.id.btn_easy);

        cardFlashcard.setOnClickListener(v -> flipCard());

        btnAgain.setOnClickListener(v -> submitGrade(SRSLogic.Grade.AGAIN));
        btnHard.setOnClickListener(v -> submitGrade(SRSLogic.Grade.HARD));
        btnGood.setOnClickListener(v -> submitGrade(SRSLogic.Grade.GOOD));
        btnEasy.setOnClickListener(v -> submitGrade(SRSLogic.Grade.EASY));
    }

    private void setupViewModels(int subjectId) {
        SubjectViewModel subjectViewModel = new ViewModelProvider(this).get(SubjectViewModel.class);
        flashcardViewModel = new ViewModelProvider(this).get(FlashcardViewModel.class);

        subjectViewModel.getSubjectById(subjectId).observe(this, subject -> {
            if (subject != null) {
                currentSubject = subject;
                String name = subject.getName();
                textSubjectTitle.setText(name != null ? name.toUpperCase() : "");
                flashcardViewModel.startSession(subject.getSubjectId(), subject.getExamDate());
            }
        });

        flashcardViewModel.getSessionCards().observe(this, cards -> {
            sessionCards = cards;
```

```
            updateUI();
        });

        flashcardViewModel.getCurrentIndex().observe(this, index -> {
            isShowingAnswer = false;
            updateUI();
        });

        flashcardViewModel.getIsSessionComplete().observe(this, isComplete -> {
            if (isComplete) {
                Intent intent = new Intent(this, SessionRecapActivity.class);
                if (sessionCards != null) {
                    intent.putExtra("mastered_count", sessionCards.size());
                }
                startActivity(intent);
                finish();
            }
        });
    }

    private void flipCard() {
        if (sessionCards == null || sessionCards.isEmpty()) return;

                    AnimatorSet flipOutSet = (AnimatorSet) AnimatorInflater.loadAnimator(this,
R.animator.card_flip_out);
                    AnimatorSet flipInSet = (AnimatorSet) AnimatorInflater.loadAnimator(this,
R.animator.card_flip_in);

        flipOutSet.setTarget(cardFlashcard);
        flipInSet.setTarget(cardFlashcard);

        flipOutSet.addListener(new android.animation.AnimatorListenerAdapter() {
            @Override
            public void onAnimationEnd(android.animation.Animator animation) {
                isShowingAnswer = !isShowingAnswer;
                updateUI();
                flipInSet.start();
            }
        });
        flipOutSet.start();
    }

    private void updateUI() {
        Integer index = flashcardViewModel.getCurrentIndex().getValue();
        if (sessionCards == null || index == null || index >= sessionCards.size()) {
            return;
        }

        Flashcard currentCard = sessionCards.get(index);
              textSessionProgress.setText(getString(R.string.session_progress_format, (index + 1),
sessionCards.size()));

        if (isShowingAnswer) {
            textCardContent.setText(currentCard.getAnswer());
            layoutGrading.setVisibility(View.VISIBLE);
            findViewById(R.id.text_tap_to_flip).setVisibility(View.INVISIBLE);
        } else {
            textCardContent.setText(currentCard.getQuestion());
            layoutGrading.setVisibility(View.GONE);
            findViewById(R.id.text_tap_to_flip).setVisibility(View.VISIBLE);
        }
    }
```

```
    private void submitGrade(SRSLogic.Grade grade) {
        Integer index = flashcardViewModel.getCurrentIndex().getValue();
        if (sessionCards != null && index != null && index < sessionCards.size()) {
            flashcardViewModel.submitGrade(sessionCards.get(index), currentSubject, grade);
        }
    }
}
```

3.4.2 Layout Declaration

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout    xmlns:android="http://schemas.android.com/apk/
res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    tools:context=".ui.StudySessionActivity">

    <TextView
        android:id="@+id/text_subject_title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textAppearance="@style/TextAppearance.Rivi.UI"
        android:textColor="@color/rivi_text_secondary"
        android:textSize="14sp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        tools:text="BIOLOGY" />

    <TextView
        android:id="@+id/text_session_progress"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textAppearance="@style/TextAppearance.Rivi.UI"
        android:textColor="@color/rivi_text_secondary"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        tools:text="2 / 5" />

    <com.google.android.material.card.MaterialCardView
        android:id="@+id/card_flashcard"
        style="@style/Widget.Rivi.CardView"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_marginTop="24dp"
        android:layout_marginBottom="32dp"
        app:layout_constraintBottom_toTopOf="@id/layout_grading"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/text_subject_title">

        <!-- Simplified for MVP, we'll use a single view and swap content -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:gravity="center"
            android:orientation="vertical"
            android:padding="32dp">
```

```
        <TextView
            android:id="@+id/text_card_content"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:textAppearance="@style/TextAppearance.Rivi.Academic"
            android:textSize="24sp"
            tools:text="What is the powerhouse of the cell?" />

        <TextView
            android:id="@+id/text_tap_to_flip"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="32dp"
            android:text="@string/tap_to_flip"
            android:textAppearance="@style/TextAppearance.Rivi.UI"
            android:textColor="@color/rivi_stroke"
            android:textSize="12sp" />
    </LinearLayout>

</com.google.android.material.card.MaterialCardView>

<LinearLayout
    android:id="@+id/layout_grading"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:visibility="gone"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    tools:visibility="visible">

    <com.google.android.material.button.MaterialButton
        android:id="@+id/btn_again"
        style="@style/Widget.Rivi.Button.Outline"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="@string/again"
        android:textColor="#F44336"
        app:strokeColor="#F44336" />

    <com.google.android.material.button.MaterialButton
        android:id="@+id/btn_hard"
        style="@style/Widget.Rivi.Button.Outline"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_weight="1"
        android:text="@string/hard"
        android:textColor="#FF9800"
        app:strokeColor="#FF9800" />

    <com.google.android.material.button.MaterialButton
        android:id="@+id/btn_good"
        style="@style/Widget.Rivi.Button.Solid"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_weight="1"
        android:backgroundTint="#4CAF50"
```

```
            android:text="@string/good"
            app:rippleColor="#20FFFFFF" />

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_easy"
            style="@style/Widget.Rivi.Button.Outline"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_marginStart="8dp"
            android:layout_weight="1"
            android:text="@string/easy"
            android:textColor="#2196F3"
            app:strokeColor="#2196F3" />

    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

3.5 Activity 3: Administration (CardManagementActivity)

Provides CRUD operations for the knowledge graph. It allows users to modify or inject new flashcards into the database.

- **Functional Logic:** Connects to the Room DAO layer directly. It uses a RecyclerView to list existing cards and provides a floating action button for additions.

3.5.1 Implementation Code

```
package me.prasad.study_app.ui;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.lifecycle.ViewModelProvider;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.textfield.TextInputEditText;

import me.prasad.study_app.R;
import me.prasad.study_app.data.entity.Flashcard;
import me.prasad.study_app.ui.adapter.FlashcardAdapter;
import me.prasad.study_app.viewmodel.FlashcardViewModel;

public class CardManagementActivity extends AppCompatActivity {

    public static final String EXTRA_SUBJECT_ID = "extra_subject_id";
```

```
    private FlashcardViewModel viewModel;
    private FlashcardAdapter adapter;
    private int subjectId;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_card_management);

        subjectId = getIntent().getIntExtra(EXTRA_SUBJECT_ID, -1);
        if (subjectId == -1) {
            finish();
            return;
        }

        setupRecyclerView();

        viewModel = new ViewModelProvider(this).get(FlashcardViewModel.class);
      viewModel.getFlashcardsForSubject(subjectId).observe(this, cards -> adapter.submitList(cards));
    }

    private void setupRecyclerView() {
        RecyclerView recyclerView = findViewById(R.id.recycler_flashcards);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));
        adapter = new FlashcardAdapter();
        recyclerView.setAdapter(adapter);

        adapter.setOnCardDeleteListener(card -> new AlertDialog.Builder(this)
                .setTitle(R.string.delete_card)
                .setMessage("Are you sure you want to delete this card?")
                .setPositiveButton("Delete", (dialog, which) -> viewModel.deleteFlashcard(card))
                .setNegativeButton("Cancel", null)
                .show());

        View fab = findViewById(R.id.fab_add_card);
        if (fab != null) {
            fab.setOnClickListener(v -> showAddCardDialog());
        }
    }

    private void showAddCardDialog() {
        View view = LayoutInflater.from(this).inflate(R.layout.dialog_add_card, null);
        TextInputEditText questionInput = view.findViewById(R.id.edit_question);
        TextInputEditText answerInput = view.findViewById(R.id.edit_answer);
        MaterialButton btnSave = view.findViewById(R.id.btn_save);
        MaterialButton btnCancel = view.findViewById(R.id.btn_cancel);

        AlertDialog dialog = new AlertDialog.Builder(this)
                .setView(view)
                .create();

        btnSave.setOnClickListener(v -> {
                                    String   question   =   questionInput.getText()   !=   null   ?
questionInput.getText().toString().trim() : "";
         String answer = answerInput.getText() != null ? answerInput.getText().toString().trim() : "";

            if (question.isEmpty() || answer.isEmpty()) {
                Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show();
                return;
            }
```

```
            // New card with default SM-2 values
             Flashcard card = new Flashcard(subjectId, question, answer, System.currentTimeMillis(),
0, 2.5f);
            viewModel.insertFlashcard(card);
            dialog.dismiss();
        });

        btnCancel.setOnClickListener(v -> dialog.dismiss());

        dialog.show();
    }
}
```

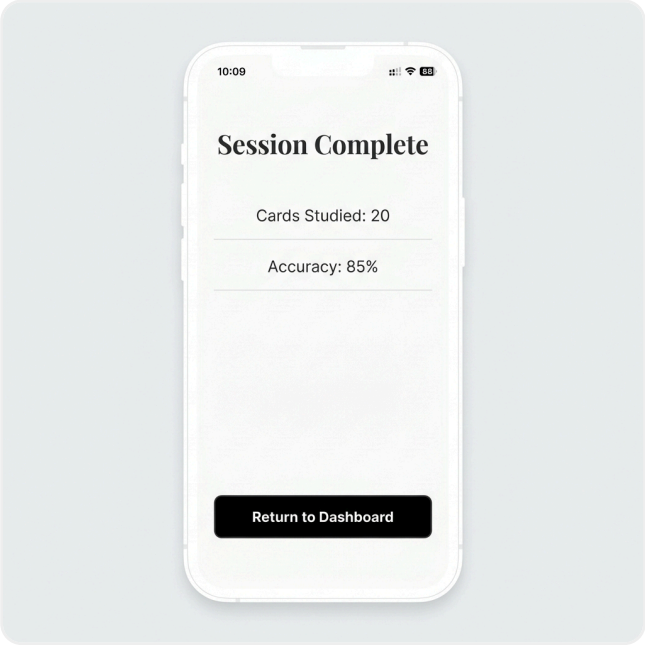
3.5.2 Layout Declaration

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout    xmlns:android="http://schemas.android.com/apk/
res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    tools:context=".ui.CardManagementActivity">

    <TextView
        android:id="@+id/text_manage_title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/manage_cards"
        android:textAppearance="@style/TextAppearance.Rivi.Academic"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recycler_flashcards"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_marginTop="16dp"
        android:clipToPadding="false"
        android:paddingBottom="88dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/text_manage_title"
        tools:listitem="@layout/item_flashcard" />

    <com.google.android.material.floatingactionbutton.FloatingActionButton
        android:id="@+id/fab_add_card"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        android:contentDescription="@string/add_new_card"
        android:src="@drawable/ic_add"
        app:backgroundTint="@color/black"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:tint="@color/white" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

3.6 Activity 4: Session Analytics (SessionRecapActivity)

Bridges the study session and the dashboard, providing positive reinforcement through data visualization.

- **Functional Logic:** Receives accuracy metrics via Intent extras and renders them into progress charts before allowing the user to return home.



3.6.1 Implementation Code

```
package me.prasad.study_app.ui;

import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.button.MaterialButton;

import me.prasad.study_app.R;

public class SessionRecapActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_session_recap);

        int count = getIntent().getIntExtra("mastered_count", 0);
        TextView textSubtitle = findViewById(R.id.text_recap_subtitle);
        textSubtitle.setText(getString(R.string.recap_subtitle, count));

        MaterialButton btnFinish = findViewById(R.id.btn_finish);
        btnFinish.setOnClickListener(v -> finish());
```

```
    }
}
```

3.6.2 Layout Declaration

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout    xmlns:android="http://schemas.android.com/apk/
res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    tools:context=".ui.SessionRecapActivity">

    <TextView
        android:id="@+id/text_recap_title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/recap_title"
        android:textAppearance="@style/TextAppearance.Rivi.Academic"
        android:textSize="32sp"
        android:textStyle="bold"
        app:layout_constraintBottom_toTopOf="@id/text_recap_subtitle"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_chainStyle="packed" />

    <TextView
        android:id="@+id/text_recap_subtitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:text="@string/recap_subtitle"
        android:textAppearance="@style/TextAppearance.Rivi.UI"
        android:textColor="@color/rivi_text_secondary"
        app:layout_constraintBottom_toTopOf="@id/btn_finish"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/text_recap_title" />

    <com.google.android.material.button.MaterialButton
        android:id="@+id/btn_finish"
        style="@style/Widget.Rivi.Button.Solid"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="48dp"
        android:padding="16dp"
        android:text="@string/back_to_dashboard"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/text_recap_subtitle" />

</androidx.constraintlayout.widget.ConstraintLayout>
```